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REGULAR ARTICLE

Hand-held computers can help to distract children undergoing painful venipuncture procedures

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ABSTRACT

Aim: Needle-related procedures can be painful for children, and distraction provides ideal pain relief in blood-drawing centres. This study assessed the effectiveness of playing a computer game during venipuncture, compared with low-tech distraction by a nurse.

Methods: We conducted this prospective, randomised controlled trial at the blood-drawing centre of a tertiary-level children's hospital in Italy. Half of the 200 children played Angry Birds on a hand-held computer while the other half were distracted by a second, specifically trained nurse who sang to them, read a book, blew bubbles or played with puppets. Pain was measured using a faces pain scale for children aged 4–7 years and a numeric scale for children aged 8–13 years.

Results: The 200 children had a median age of eight years. Children reported significant pain in 16 cases (16%) in the hand-held computer distraction group and in 15 cases (15%) in the nurse-led low-tech distraction group ($p = 0.85$). The procedural success rate at the first attempt was not different in the two groups.

Conclusion: Playing a game on a hand-held computer meant that only one in six children reported pain during venipuncture, but it was not superior to being distracted by nurses.

INTRODUCTION

Pain is a subjective experience that comprises sensory, emotional, cognitive and behavioural components (1), and recurrent pain stimuli may interfere with children's emotional development and behaviour (2,3).

Needle-related procedures, including venipuncture, are among the most frequently performed painful procedures on children, both in and out of hospital.

Research has shown that up to 60% of children reported pain and distress during venipuncture (4), indicating that both pain and distress management is warranted.

International guidelines recommend behavioural and environmental support, followed by pharmacological and physical treatment (5).

There is strong evidence that psychological interventions are useful when it comes to relieving pain and distress during needle procedures (6), and distraction is the most studied technique. Distraction is a cognitive strategy that aims to divert the child's attention from a noxious stimulus and it plays a central role because it interferes with the neuronal activity associated with pain perception and with cognitive processing of afferent pain stimuli (7).

Active distraction encourages the child to do something different during painful procedures, such as playing or blowing, while passive distraction redirects the child's attention to visual or auditory stimuli, using toys, songs, movies or blowing bubbles (7).

A meta-analysis showed strong evidence supporting the efficacy of distraction on self-reported pain during needle procedures. However, it noted strong variability in the distraction methods that were used and studies comparing different types of distraction are therefore still needed (6).

Abbreviations
FPS-R, Faces pain scale-revised; IQR, Interquartile range.

Key notes
- Distraction is a useful intervention for venipuncture pain in children, but there is no evidence of which distraction technique is the most useful.
- We performed a randomised controlled trial to compare the efficacy of a hand-held computer game with nurse-led distraction in a blood-drawing centre.
- Significant procedural pain was reported by 16% of children in the hand-held computer distraction group and by 15% in the nurse distraction group.

When it comes to the actual distraction techniques, there is an increasing interest in the use of new technology, such as electronic games and virtual reality (8–10).

Some reports have drawn attention to the effectiveness of hand-held computers as a form of distraction during painful procedures in children, but there are some gaps that still need to be addressed (11–13).

A blood-drawing centre is a particular setting in which many procedures must be performed in a limited time. Patients usually arrive without pharmacological premedication and are discharged immediately after the procedure. In this context, distraction is an ideal pain relief tool.

To date, only a few studies have used hand-held computers to provide a distraction and relieve pain during needle procedures (12,13) and none were carried out in the setting of a blood-drawing centre. Moreover, no study has compared the effectiveness of distracting a paediatric patient by getting them to play a game on a hand-held computer with the low-tech distraction provided by specially trained nurses. The aim of our study was to compare the effectiveness of these two techniques on pain relief during venipuncture at a blood-drawing centre.

PATIENTS AND METHODS

This study was an open, randomised control trial with two parallel arms. It was conducted between March and June 2013 at the blood-drawing centre of the Institute for Maternal and Child Health IRCCS Burlo Garofolo of Trieste, Italy, a tertiary-level children hospital.

Our blood-drawing centre performs routine or nonurgent blood tests on paediatric outpatients. Patients are referred to the centre by family physicians and are feeling relatively well, usually without fever or pain. Blood samples are mainly performed to rule out blood diseases, such as anaemia or leucopenia, as well as iron deficiency, coeliac disease, inflammatory or chronic diseases and endocrinological diseases. Nurse-led low-tech distraction is the standard strategy used for pain management at our blood-drawing centre.

The eligible subjects were children from four to 13 years of age who came to the blood-drawing centre to have a blood sample collected by venipuncture. The exclusion criteria were as follows: a patient history of epilepsy, use of topical, enteral or parenteral analgesics up to eight hours before blood drawing, the inability to have venipuncture performed on their hand or arm, the presence of cognitive impairment or the inability to report pain verbally.

In line with the evidence that a suitable environment decreases the distress experienced by children during painful procedures (14), all eligible children and their parents were taken to a comfortable, child-friendly quiet room before procedure. The room was away from the blood-drawing centre waiting room, and the parents were invited to hold or stay close to their children during venipuncture in order to decrease the children's distress (15).

The eligible children were randomly allocated to be actively distracted using an ASUS Transformer Pad TF300T hand-held computer (ASUSTek Computer Inc, Taipei, Taiwan), or to receive the nurse-led low-tech distraction performed by a specifically trained nurse.

The children in the computer group played Angry Birds (Rovio Entertainment Ltd, Espoo, Finland), an easy game in which they had to shoot birds into the pig's fortresses. Children started to play the game three minutes before the procedure and then continued for a maximum of three minutes after the procedure. The children were able to play the game with one hand, so it was easy for them to continue playing while the blood was being drawn.

In the active hand-held computer distraction group, only the nurse performing the procedure was present in the room, with a second nurse being immediately available if help was needed. Children in the nurse-led distraction group received various kinds of conventional distractions, starting three minutes before the blood sample. These included nurses singing a song, reading a book, blowing bubbles and performing a puppet show. The technique that most engaged the child was continued during the procedure. In this group, one nurse was in charge of the needle procedure and a second one performed the distraction.

Trained nurses were incharge of the whole process. All nurses working at the blood-drawing centre had received standard training in pain and distress management that focused on distraction techniques. These included dedicated training sessions, led by a paediatric psychologist, which consisted of lectures and practical exercises.

All the study nurses had specific experience of caring for children during needle procedures and had worked in the blood-drawing centre for at least two years.

The equipment was hidden from the patients' view, and the venipuncture procedure only started when the child had had a chance to settle in, not as soon as they arrived.

A randomisation list was generated using a computer-based method by an independent epidemiologist at the Clinical Epidemiology and Public Health Research Unit at our Institute. Allocation concealment was guaranteed using sealed, consecutively numbered opaque envelopes, each containing the allocation group. After obtaining each informed consent, the nurse opened the lowest numbered envelope available and assigned the patients to the corresponding group.

The primary study outcome was self-reported procedural pain score in the two groups. Children from four to seven years of age recorded pain using the Faces Pain Scale-Revised (FPS-R) (16). The scale contains six faces aligned in increasing pain intensity at equal intervals, and children had to indicate the face corresponding to their pain. This information was then transformed into a numerical value from zero for no pain to 10 for maximum pain.

Children from eight to 13 years recorded pain with a numeric rating scale, from zero to 10. For both scales, values exceeding four were considered significant pain.

The scale was given to the child within five minutes of the painful procedure.

The secondary outcomes were the venipuncture success rate at the first attempt in the two groups and the number and type of adverse events in the two groups.

The following variables were also collected for each child: age, gender, site of procedure, number of attempts, history of previous needle procedures and the presence of a chronic disease.

The operator performing the procedure filled out the data collection form.

The study received approval from the Independent Bioethic Committee of the Institute. All the children's parents provided written informed consent for participation. The trial was registered with ClinicalTrial.gov (NCT02614391).

Statistical analyses

The study was designed as a superiority trial. Data from the literature allow us to estimate that children who received passive conventional distraction would report a mean pain value of around 2.5 with a standard deviation ranging from 1.5–2. Based on this, we hypothesised that in order to demonstrate an improvement in the pain score of 0.8 in the intervention group, 200 children – 100 for each group – were needed to carry out the study, with $\alpha = 0.05$ and $\beta = 0.20$.

Analyses were carried out according to the intention-to-treat principle. Continuous variables are reported as medians, and interquartile ranges (IQR) with categorical data reported as numbers and percentages. Differences between the groups were evaluated with the chi-squared test or Fisher's exact test for categorical variables and with the nonparametric Mann–Whitney *U*-test for continuous variables. A non-normal distribution of data was shown both visually and with the Kolmogorov–Smirnov test. Data were analysed with SPSS software, version 21.0 (IBM Corp, Armonk, New York, USA). A p value of <0.05 was considered statistically significant.

RESULTS

We assessed 373 children for eligibility. Of these, 16 declined to participate and 157 were excluded because they did not fulfil the study criteria: 128 had used topical anaesthetics, 21 had epilepsy, five reported analgesic use less than eight hours before enrolment and three had cognitive impairment. This meant that 200 children entered the study: 100 were randomised to the hand-held computer group and 100 to the nurse-led low-tech distraction group (Fig. 1). One child randomised to the nurse-led low-tech distraction group received hand-held computer distraction in error, but following the intention-to-treat principle, this subject was considered to belong to his respective randomisation group.

No subject was lost to follow-up. The baseline characteristics of the two groups were similar (Table 1).

The self-reported procedural pain scores were similar in the two groups (median 1.0, IQR 0.0–3.8 and 1.0, IQR 0.0–2.0, respectively, p = 0.50). Children reported significant pain in 16 cases (16%) in the hand-held computer group and in 15 cases (15%) in the nurse-led low-tech distraction group (p = 0.85) (Table 2). Additional information on procedural pain score values are reported in the supplementary table (S1). Most procedures were successful at the first attempt in both groups, 99% in the hand-held computer group and 97% in the nurse-led low-tech distraction group (p = 0.62) (Table 2). No parent refused to hold or stay close to the child during procedure. No adverse events were reported in the two groups.

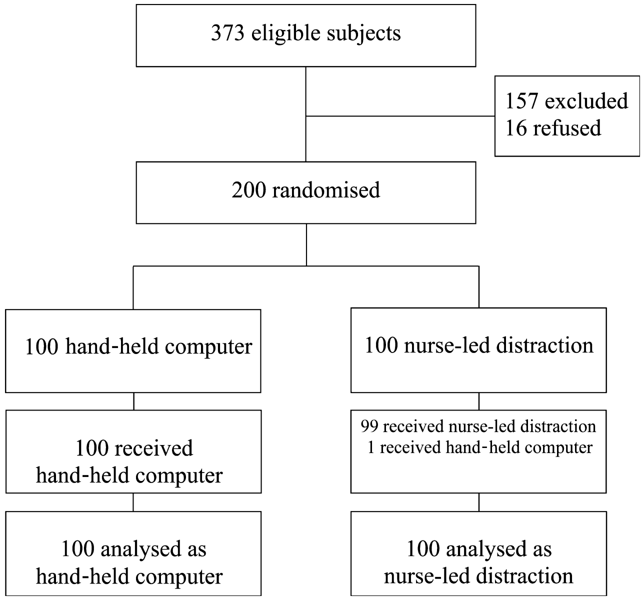

Figure 1 Study algorithm (intention-to-treat analysis).

Table 1 Patients' characteristics

	Hand-held computer (n = 100)	Nurse-led low-tech distraction (n = 100)
Male, sex, number (%)	53 (53.0%)	45 (45.0%)
Age, median (IQR)	8.6 (6.1–11.0)	8.9 (6.0–10.8)
Venipuncture, number (%)	100 (100.0%)	100 (100.0%)
Procedure at the cubital fossa, number (%)	100 (100.0%)	100 (100.0%)
Subject with chronic disease, number (%)	12 (12.0%)	19 (19.0%)

Table 2 Main study outcomes

	Hand-held computer (n = 100)	Nurse-led low-tech distraction (n = 100)	p value
Pain score, median (IQR)	1.0 (0.0–3.8)	1.0 (0.0–2.0)	0.50
Pain score >4, number (%)	16 (16.0%)	15 (15.0%)	0.85
Success at the first attempt, number (%)	99 (99.0%)	97 (97.0%)	0.62

DISCUSSION

The results of this study did not show statistically significant differences between hand-held computer distraction and nurse-led low-tech distraction performed by trained nurses in pain relief in a blood-drawing centre. Both the interventions were effective in about 85% of children.

Strong evidence has been published that distraction is a useful psychological intervention for venipuncture-related pain and distress in children (6,7), but the significant variability in distraction interventions makes difficult to identify which technique is the most useful for each patient and setting (1).

Trials have now started distinguishing between active and passive and high-tech and low-tech distraction. Active distraction techniques engage more senses and require manipulation of the environment and/or, problem solving. For this reason, it has been hypothesised that these techniques effectively distract children better than passive distraction techniques. Nevertheless, evidence has been discordant, as several studies have not confirmed this superiority (17–19).

The efficacy of electronic games in pain relief during painful procedures has been established (20–22), but comparisons of efficacy between this specific kind of active distraction and passive distraction techniques during needle procedures have led to conflicting results. MacLaren et al. (23) showed that playing a video game was less effective in providing pain relief than watching television. Another study showed that playing video games did not lead to any improvements in analgesia during venipuncture in children treated with local anaesthetic (9).

A hand-held computer is a reusable tool, with a relatively low cost, that offers a technological-based active distraction. A number of games are available that can be played with just one hand, which means that children can continue to play them during needle procedures.

Little is known about their usefulness in providing pain relief during needle procedures.

One trial showed that parents reported their children felt less pain when they were distracted by an iPad (13), while another study did not show any effect in decreasing pain when children received an injection (12).

To the best of our knowledge, this study was the first that used a hand-held computer as a form of distraction during venipuncture in a blood-drawing centre. It compared the effectiveness of the hand-held computer distraction with low-tech distractions performed by nurses, specially trained in pain management and used to providing high-quality children's care. The results showed no difference in self-reported procedural pain between these two different kinds of distraction.

Moreover, hand-held computer distraction did not influence the procedural success rate at the first attempt and procedural adverse events.

The procedural success rate at the first attempt in both groups was extremely high: 99% in the hand-held computer group and 97% in the nurse-led distraction group. This may be explained by the fact that patients with chronic diseases or patients with known problems with vascular access do not normally attend the blood-drawing centre. Furthermore, children who had to undergo repeated needle procedures frequently arrived with a pharmacological topical premedication, and therefore, they were not included in the study. In any case, the number of children with a chronic disease did not differ significantly between the two study groups.

The main advantage of hand-held computer distraction is that it does not require any specific training so that operators without specific paediatric assistance skills may use this tool.

This does not mean that a hand-held computer can replace an experienced and trained nurse (24), but it may provide a valid alternative to conventional passive distraction techniques.

The study results were strengthened by the significant number of children enrolled and by the high level of experience and motivation of the staff employed to perform the passive low-tech distraction.

The study has some limitations. It was not blinded, but blinding was not possible given the nature of the techniques we used, and we only recorded pain scores without investigating children's anxiety and stress. However, we aimed to design a simple pragmatic trial with a substantial outcome.

Playing with the hand-held computer was not compared to a single passive low-tech distraction technique, and the data collection form did not specify which technique was used case by case. For these reasons, the study was not able to assess whether playing with the hand-held computer was more or less effective than a single, passive, low-tech distraction technique.

CONCLUSION

Hand-held computer distraction appeared to be an effective pain management strategy in a blood-drawing centre. It provided the same support as conventional, passive low-tech distraction techniques performed by trained, skilled nurses. In this setting, hand-held computers may be a valid care option if there are limited resources or a shortage of trained staff.

CONFLICT OF INTEREST

The authors do not have any conflict of interest to declare.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Table S1 Additional information on pain score values by study groups.